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#include <LPC21xx.H>

#include "header.h"

#include<stdio.h>

void uart0_init(unsigned int baud)
{
    unsigned int result=0,pclk;
    int a[]={15,60,30,15,15};
    pclk=a[VPBDIV]*1000000;
    result=pclk/(16*baud);
    PINSEL0|=5;    // txd0 & rxd0

    U0LCR=0X83;          // DLAB=1

    U0DLL=result & 0x00ff;    // UART 0 INTILIZATION

    U0DLM=(result>>8)&0xff;

    U0LCR=3;          //DLAB=0
}

#define THRE ((U0LSR>>5)&1)          // cheak TX FLAG

void uart0_tx(unsigned char data)
{
    U0THR=data;
    while (THRE==0);
}

#define RDR (U0LSR&1)    // cheak recive flag

unsigned char uart0_rx(void)
{
    while(RDR==0);
    return U0RBR;
}

```

```
void uart0_tx_string(char *ptr)
{
    while(*ptr)
    {
        U0THR=*ptr;
        while(THRE==0);
        ptr++;
    }
}

void uart0_tx_integer(int temp)
{
    char a[10] = {0};
    sprintf(a,"%d",temp);
    uart0_tx_string(a);
}

void uart0_tx_float(float temp)
{
    char a[10] = {0};
    sprintf(a,"%f",temp);
    uart0_tx_string(a);
}
```